

Helix OTA — Session Resumption (canonical, always-current)

Field	Value
Revision	13
Last modified	2026-07-26T20:25:00Z
Status	active — production completion guide published; 9-doc comprehensive step-by-step plan at docs/production/completion/;
Standard path	docs/RESUMPTION.md (this file)

SHORT — paste this first sentence into a fresh session

Helix OTA 1.0.0 released on main. 72/73 gaps closed. 9-doc comprehensive production completion guide at docs/production/completion/00_MASTER_INDEX.md — read THAT FIRST. It enumerates every remaining step (Stages A–I), 12 operator decisions, 8 danger zones, and the full critical path from current state → production deployment. Run `git fetch --all --prune`. Start at docs/production/completion/01_OPERATOR_DECISIONS.md to resolve blocking decisions.

FULL — detailed resumption block

0. Read FIRST (in order)

1. docs/production/completion/00_MASTER_INDEX.md — THE authoritative production completion guide (NEW — 9-doc series, Stages A–I, every remaining step enumerated)
2. docs/research/main_specs/CONTINUATION.md — the live work-state handoff

3. `.remember/remember.md` — session memory (updated this session)
4. `docs/research/production_planning_20260726/ANALYSIS.md` — production readiness planning document (§11.4.172)
5. `docs/research/production_planning_20260726/gap_tracker.csv` — 73-gap inventory
6. Then run: `git fetch --all --prune`

1. Exact live-state anchors

- **HEAD commit:** Run `git rev-parse HEAD` and `git log --oneline -1` to determine current HEAD
- **Branch:** `main`
- **Release tag:** `helix_ota-1.0.0`
- **Remote status:** Run `git fetch --all --prune && git status` to verify
- **Upstreams (4):** `github` (`HelixDevelopment/helix_ota`), `gitlab` (`helixdevelopment1/helix_ota`), `gitflic` (`helixdevelopment/helix_ota`), `gitverse` (`helixdevelopment/helix_ota`)

2. Phase Completion Status

Phase	Status	Notes
Setup (env, deps, submodules)	Done	Go 1.26.2, Gradle 9.5, Kotlin 2.3.20
Foundation (US0 — scaffolding)	Done	Project structure, build system, gates
US1 — API + Auth	Done	Gin REST API, JWT auth, RBAC, OpenAPI
US2 — Store + Transport	Done	PostgreSQL store, MinIO/S3, HTTP/3 + Brotli
US3 — Rollout + Orchestration	Done	Staged rollout, device emulator, telemetry
US4 — Production Readiness	Done	Planning doc, ADR acceptance, velocity tracking, gap closure
US5 — Rate-limit/ Security Hardening	Done	Rate-limiter config, security probes extended
US6 — Fuzz/Chaos Testing	Done	

Phase	Status	Notes
		Fuzz inputs, stress+chaos across components

3. Production Readiness State — POST-1.0.0

- **Gap tracker:** docs/research/production_planning_20260726/gap_tracker.csv — 73 gaps, 72 Closed, 1 hardware-gated (Tier-3: real RK3588), 0 Queued
- **Planning doc:** docs/research/production_planning_20260726/ANALYSIS.md — §11.4.172 compliant
- **Velocity tracking:** scripts/track_velocity.sh — appends to docs/research/production_planning_20260726/velocity.tsv
- **ADRs:** All 5 formally Accepted (2026-07-26):
 - ADR-0001: hawkBit front-runner, AOSP-native fallback
 - ADR-0002: MVP plain signing; TUF server-side 1.0.1+
 - ADR-0003: Modular monolith with extractable seams
 - ADR-0004: HTTP/3 + Brotli, 2-class compression
 - ADR-0005: Full payload MVP; AOSP incrementals post-MVP
- **Workable items:** 16 completed in cleanup waves, synced to DB. 4 items hardware-gated (RK3588 board required): OTA-004, OTA-013, OTA-017, OTA-025 — all documented with unblock conditions.
- **Carrier lockstep:** CLAUDE.md / AGENTS.md / GEMINI.md / QWEN.md refreshed and in sync per §11.4.157.

4. Evidence Paths

Scope	Path
qa-results/ (submodules)	submodules/challenges/qa-results/, submodules/containers/qa-results/
docs_chain qa-results	docs_chain/qa-results/ docs_chain/
Stability report	docs/qa/STABILITY_REPORT.md
Carrier checkpoint	qa-results/pending-final/ (CLAUDE.md, AGENTS.md, GEMINI.md, QWEN.md)

5. Device States

- **RK3588 / Orange Pi 5 Max:** No current device connected. Tier-3 hardware testing deferred per docs/design/EMULATED_DEVICE_TESTING.md.
- **Emulator (podman):** Tier-1 container e2e PROVEN on main. Full lifecycle + multi-device fleet + recall/recovery all GREEN.
- **Cuttlefish (Linux+KVM):** Honest SKIP on this macOS host. Design ready at docs/design/CUTTLEFISH_TIER2.md. Needs Linux+KVM host.

6. Binding Constraints (unchanged)

- **Anti-bluff §11.4:** Every PASS carries positive captured evidence
- **No force-push §11.4.113:** Strictly forbidden on every repo/submodule
- **Podman-only on this host:** Use containers submodule; never host-direct emulator/adb/qemu
- **Tier-2 Android A/B host-gated:** Needs Linux+KVM for Cuttlefish
- **Commit/push discipline:** Commit + push to ALL 4 upstreams; pushes may run detached

7. Recent Migrations

- **OTA-031** (2026-07-26): Migrated OTA-003 from Issues.md to Fixed.md (was duplicate — OTA-003 already archived in Fixed.md). Removed the stale entry from Issues.md.
- **Carrier lockstep refresh** (2026-07-26): Metadata tables added to CLAUDE.md, AGENTS.md, GEMINI.md; QWEN.md created; all four locked to helix_ota-1.0.0 per §11.4.157.
- **DB sync** (2026-07-26): 16 pending workable items synced to workable_items.db.
- **Production completion guide** (2026-07-26): 9-doc comprehensive step-by-step guide at docs/production/completion/. 40+3 exports (HTML+PDF+DOCX). All files committed and pushed to 4 upstreams.

8. Immediate NEXT (post-1.0.0 — follow the completion guide)

1. **Read docs/production/completion/00_MASTER_INDEX.md** — this is THE plan
2. **Resolve Stage A** (12 operator decisions in 01_OPERATOR_DECISIONS.md) — these BLOCK all major work
3. **Run Stage B** (server hardening) in parallel with operator decisions

4. **When A-01/A-02 resolved, start Stage C** (multi-tenant Accounts — XL effort, critical path)
5. **When hardware available, start Stage E** (device-side completion)
6. **When A-07..A-11 resolved, start Stage F** (deployment infrastructure)
7. **Stage H-09** (§11.4.185 manual QA) is the FINAL gate — nothing is “production” until it passes
8. Regular main → feature merge cadence per §11.4.188